



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-5

Missionaries and Cannibals Problem

Aim

To write a Python program to solve the Missionaries and Cannibals Problem.

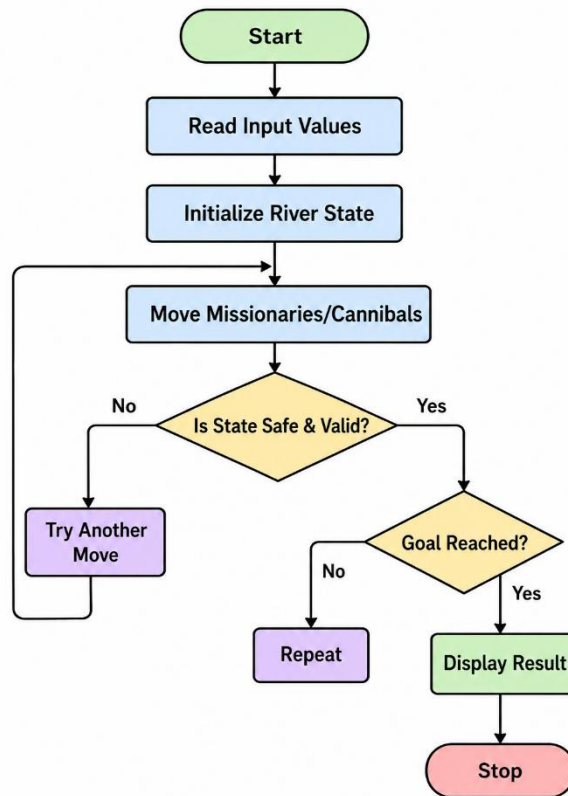
Objective

To safely move all missionaries and cannibals across the river.

Algorithm

1. Start.
2. Read the number of missionaries and cannibals.
3. Initialize the left bank with all missionaries and cannibals.
4. Move one or two people across the river.
5. Ensure missionaries are never outnumbered by cannibals on either bank.
6. Repeat until everyone reaches the opposite bank.
7. Display the solution.
8. Stop.

Flowchart



Code

```
m = int(input("Enter number of Missionaries: "))
```

```
c = int(input("Enter number of Cannibals: "))
```

```
print("\nInitial State:")
```

```
print(f'Left Bank : M={m}, C={c}')
```

```
print("Right Bank: M=0, C=0")
```

```
if m == c and m > 0:
```

```
    print("\nSolution Steps:")
```

```

print("Move 2 Cannibals")
print("Move 1 Cannibal back")
print("Move 2 Missionaries")
print("Move 1 Missionary and 1 Cannibal back")
print("Continue similar safe moves until all cross.")

```

else:

```

print("\nThis simple demonstration works for equal numbers of
missionaries and cannibals.")

```

Output

```

1  m = int(input("Enter number of Missionaries: "))
2  c = int(input("Enter number of Cannibals: "))
3
4  print("\nInitial State:")
5  print(f"Left Bank : M={m}, C={c}")
6  print(f"Right Bank: M=0, C=0")
7
8  if m == c and m > 0:
9      print("\nSolution Steps:")
10     print("Move 2 Cannibals")
11     print("Move 1 Cannibal back")
12     print("Move 2 Missionaries")
13     print("Move 1 Missionary and 1 Cannibal back")
14     print("Continue similar safe moves until all cross.")
15 else:
16     print("\nThis simple demonstration works for equal numbers of missionaries and cannibals.")

```

Enter number of Missionaries: 3
Enter number of Cannibals: 3

Initial State:
Left Bank : M=3, C=3
Right Bank: M=0, C=0

Solution Steps:
Move 2 Cannibals
Move 1 Cannibal back
Move 2 Missionaries
Move 1 Missionary and 1 Cannibal back
Continue similar safe moves until all cross.

Result

The Missionaries and Cannibals problem was executed successfully.

Conclusion

The program demonstrates a safe sequence of moves to solve the Missionaries and Cannibals problem.